

Mining Atlanta's Restaurant Reviews

Text Mining Analysis of Atlanta Restaurant Reviews

Nova Information Management School

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1. Introduction

These days, the internet is overflowing with user generated content, creating a massive demand for tools that can turn unstructured text into real insights. In the business world, customer reviews are a perfect example because while a star rating gives you a snapshot, the actual text provides the 'why' that's often hidden behind the score.

This report explores how text mining can be applied to restaurant reviews. Our findings show that the words customers use, even when analyzed separately from their ratings, are incredibly accurate predictors of satisfaction. Often, the text reveals nuances that a rating alone misses. Ultimately, text mining proves to be an essential tool for turning mountains of raw feedback into clear, actionable insights.

2. Literature Review

To address the limitations of numerical ratings, which often suffer from extremity bias, research suggests that text based sentiment analysis provides a more balanced and insightful measure of customer satisfaction [\[2\]](#). Recent studies indicate that while Naive Bayes offers a computationally efficient baseline, ensemble methods like Random Forest and optimized Support Vector Machines are better suited for the complexities of multi-class classification and fine grained sentiment tasks [\[3\]](#), [\[4\]](#). By prioritizing these algorithms, we can better account for the nuanced language found in restaurant reviews, going beyond simple positive/negative labels to capture the full range of what customers really think.

Effective feature extraction is shown to be equally critical, evidence shows that TF-IDF weighting combined with bigrams and trigrams is necessary to preserve the

meaning of multi word phrases, such as specific dish names or negated sentiments, which unigrams frequently overlook [4]. Furthermore, a standard preprocessing pipeline involving tokenization and lemmatization is essential for grouping related concepts, though care must be taken with frequency filtering to ensure that rare but informative cuisine-specific terms are not discarded [1]. Overall, these results give us some solid guidelines for choosing our models. We'll keep them in mind as we move forward with the project to make sure the insights we pull from the reviews truly capture what customers are experiencing, going well beyond what the star ratings tell us on their own.

3. Data Understanding

3.1 Dataset Overview

The dataset comprises 53,566 Google Maps reviews scraped from 295 Atlanta restaurants in October 2023. Table 1 summarizes the dataset characteristics.

Table 1: Dataset Overview

Feature	Description
Total Reviews	53,566
Unique Restaurants	295
Missing Text Values	0 (0.00%)
Duplicate Reviews	99 (0.18%)
Cuisine Categories	19 unique types
Star Ratings	1-5 scale

The distribution of star ratings is heavily skewed, with an overwhelming majority of reviews being 5-star (62%). As illustrated in [Figure 1](#), a smaller secondary spike occurs at 1-star (13%), indicating polarized user sentiment. Middle ratings (2-4 stars) are significantly less common, collectively accounting for only 25% of reviews. This polarization will lead to challenges in sentiment prediction later on at intermediate rating levels, as the model will have fewer examples of neutral or moderate feedback to learn from.

3.2 Text Characteristics

Prior to textual analysis, the language of each review was automatically detected to determine if we had to do any translation to ensure data consistency and standardization across the data. Language detection identified that **93%** of reviews are in English, with **7%** requiring translation from Spanish, Somali, and other languages, as seen in [Figure 2](#). Non-English reviews were translated using Google Translate API to ensure corpus consistency.

The majority of reviews are concise, with a median length well below 100 words. The distribution of review length is heavily right-skewed, shown in [Figure 3\(a\)](#); while most reviews are short, a small number of extreme outliers reach lengths exceeding 600 words, with the longest reaching 751 words. There is a clear relationship between review length and star rating: reviews with lower ratings tend to be substantially longer, with the average word count of a 5-star review being less than half that of a 1-star review, as seen in [Figure 3\(b\)](#). This suggests dissatisfied customers provide more detailed explanations, whereas positive feedback is often brief, which aligns with expectations.

3.3 Lexical Analysis

We conducted exploratory analysis of frequent unigrams and bigrams to gain initial insights into the content of the corpus. After augmenting the default stopword list with domain specific terms to refine the results, the analysis revealed that words like **"great"** and **"good"** are the most frequent non-stopwords, shown in [Figure 4\(a\)](#), which aligns with the expected positive sentiment skew of the dataset. Beyond general sentiment, the emergence of specific food based tokens such as **"chicken"** and **"pizza"**, as well as service related bigrams seen in [Figure 4\(b\)](#) like **"the staff"** and **"customer service"**, provides us a clear indication that the corpus contains sufficient thematic detail to support robust feature extraction for modelling later on. We also formed a word cloud in [Figure 4\(c\)](#) to give us a better overview of the overall sentiment of the reviews.

3.4 Category Distribution And Class Imbalance

An exploratory analysis of the restaurant categories reveals a significant class imbalance within the dataset as seen in [Figure 5](#). As shown in the category distribution, **Mexican restaurants** represent the largest share with 8,940 entries, followed by **Fast Food**, **Italian**, and **American** restaurants, which each exceed 5,000 reviews. In contrast, categories such as **Bar & Grill**, **Barbecue**, and **Thai** restaurants appear much less frequently. This uneven distribution suggests the model may naturally favor majority classes; therefore, we must account for this imbalance during the modeling phase to ensure stable performance across all 19 cuisine types.

4. Preprocessing

Initial exploration revealed data quality challenges typical of user generated content: inconsistent formatting, multilingual reviews, informal language with emojis and excessive punctuation, and significant variation in review length. A key insight during experimentation was that preprocessing requirements varied across tasks, sentiment analysis required preservation of features like capitalization and punctuation that carry sentiment signals, while cuisine classification benefited from aggressive

normalization to reduce vocabulary and focus on distinct words. As such, for this project we approached the preprocessing in a more deterministic way, tailoring it to each specific task, rather than relying on a one size fits all approach.

4.1 Data Cleaning And Transformation

We created a reusable MainPipeline class to help streamline preprocessing and avoid repetitive code across tasks. The class takes configurable parameters, enabling different cleaning steps to be toggled on or off depending on the sections needs:

- **URL removal:** Regular expressions removed embedded links
- **Newline normalization:** Newline characters replaced with spaces
- **Emoji removal:** Unicode emoji characters stripped when enabled
- **Hashtag/mention handling:** Configurable removal of # and @ symbols (with option to retain the words)
- **Punctuation removal:** Special characters and punctuation stripped
- **Diacritic normalization:** Characters like "ä" and "é" converted to ASCII equivalents using Unicode NFKD normalization
- **Contraction expansion:** "n't" → "not", "m" → "am", "s" → "is"
- **Stopword removal:** NLTK English stopwords plus custom domain-specific stopwords
- **Lemmatization:** WordNetLemmatizer applied across all POS tags (noun, verb, adjective, adverb, satellite adjective)
- **Lowercase conversion**
- **Minimum token length filtering:** Tokens below configurable length threshold removed

Duplicate review removal was handled separately during data loading, with 99 duplicate reviews identified and removed.

5. Information requirements: Sentiment analysis

5.0 Preprocessing

For sentiment analysis using VADER and TextBlob, minimal preprocessing was applied to preserve sentiment signals. Stopwords, punctuation, and case were retained as these are critical for lexicon based methods, VADER interprets capitalization as emphasis and uses punctuation to modify intensity scores. Emojis were preserved as they carry sentiment information. The only cleaning applied was replacement of newline characters with spaces, removal of URLs, and removal of isolated consonants for consistent tokenization.

For sentiment analysis, reviews containing 3 sentences or less were removed from the analysis, thus 4410 reviews were discarded for the assessment of sentiment, as minimal reviews provide no contextual understanding for the models to extract sentiment.

5.1 Feature extraction

Two scoring approaches were implemented for each lexicon based method:

- **Overall scoring:** Sentiment calculated on the complete review text as a single document
- **Average sentence scoring:** Sentiment calculated per sentence, then averaged across the review

For VADER, four metrics were extracted: compound polarity (normalized aggregate score), positive proportion, neutral proportion, and negative proportion. For TextBlob, polarity (ranging -1 to +1) and subjectivity (ranging 0 to 1) scores were extracted.

To enable direct comparison with star ratings, VADER compound scores were normalized to a 1-5 scale using MinMaxScaler.

5.2 Sentiment Analysis models

5.2.1 VADER

We employed the VADER model in our analysis, thus we considered the following predictors for modeling star ratings:

1. **Overall VADER score:** a normalized sentiment score evaluated per review
2. **Average VADER score:** normalized sentiment score evaluated per sentence
3. **Positive-Negative VADER differential:** the difference between positive and negative scores, representing the net sentiment, for both review / sentence based evaluation

5.2.2 Textblob

In contrast, we also tested TextBlob for sentiment extraction, similar to VADER, averages per sentence and scores for the whole corpus. However, TextBlob exhibited poor predictive performance for this dataset. A closer examination revealed that its scores displayed high subjectivity and less consistency [Figure 6](#). across all ratings reviews, making it unreliable for modeling star ratings in our context.

5.2.3 Pretrained Model

A pretrained emotion detection model was also applied to classify each review into emotion categories, providing an alternative sentiment signal beyond polarity scores and a further comparison for the lexicon-based models.

5.2.3 Evaluation

The results indicate that overall VADER better captured the general sentiment trends in reviews effectively, providing meaningful signals that correlate with the star ratings assigned by users, explaining almost 70% of the variance. From deep exploration, one could conclude that the VADER is more confident in high-rated reviews and performs not so accurately in low-rated ones [Figure 7](#).

In the end, the pretrained model was not a reliable source of comparison, since it standardizes the reviews and classifies it mostly on negative or positive reviews, not being able to extract significant understanding of the ratings structure [Figure 8](#).

One could also insight that longer reviews need to be better considered and carefully evaluated for analysis of sentiment, as models may not capture the true meaning of those reviews. As for the rest, sentiment and furthermore comprehension can be successfully extracted from the rest of the corpus. Notably, the quality of service provided by waitstaff emerged as a critical factor influencing review polarity, ranging from friendly to rude interactions [Figure 9](#). Time delays were identified as the primary area for improvement, whereas the ambience of the establishments is consistently positively mentioned by customers [Figure 10](#). A deeper examination at the restaurant level revealed that fast-food establishments received comparatively lower scores [Figure 11](#).

6. Cuisine classification

6.0 Preprocessing and Feature Extraction

For cuisine classification, a different preprocessing strategy was required compared to sentiment analysis. The MainPipeline class was configured with more aggressive text normalization: stopword removal, lemmatization, lowercase conversion, punctuation removal, and a minimum word length filter of 2 characters. This approach reduces vocabulary size and focuses the model on distinctive content words rather than common function words.

A key innovation in our approach was the application of a pretrained Named Entity Recognition (NER) model (InstaFoodRoBERTa-NER) to identify reviews containing food related content. The hypothesis was that reviews mentioning specific dishes or ingredients would provide stronger signals for cuisine classification than reviews focused on service, ambience, or other non-food aspects. The NER model identified food entities with a confidence threshold of 0.7, resulting in two datasets:

Dataset	Reviews	Percentage
All Reviews	50,834	100%
Food-Only Reviews	23,466	46.2%

Retention rates varied by cuisine, with Pizza (65.4%) and Hamburger (68.4%) restaurants showing the highest food mention rates, while Breakfast (36.8%) and Fast Food (37.3%) showed the lowest.

TF-IDF vectorization was applied with hyperparameters optimized through grid search. The final configuration used `max_features=10,000`, `ngram_range=(1,1)`, and `min_df=2`.

6.1 Model Selection

Four classification algorithms were evaluated on both datasets: Logistic Regression, Naive Bayes, Random Forest, and Linear SVM. All models used class weighting to address the significant class imbalance across the 19 cuisine categories. Initial evaluation with default parameters showed Logistic Regression and Linear SVM achieving the highest performance across both datasets, with Logistic Regression selected for hyperparameter tuning due to its interpretability and marginally better F1 scores, as seen in [Figure 12](#).

Grid search with 5-fold stratified cross-validation was performed on the food-only dataset, optimizing TF-IDF parameters (`max_features`, `ngram_range`, `min_df`) and the regularization parameter `C`. The best configuration achieved a cross-validation F1 score of 0.666, representing a >5% improvement over default parameters.

6.2 Evaluation And Results

[Figure 13](#) presents the per-class F1 scores comparing the all reviews model against the food only model, both using optimized hyperparameters.

The food only filtering improved classification performance for 18 of 19 cuisine categories, with an average F1 score improvement of +0.179. Cuisines with distinctive dishes (Indian, Thai, Japanese) showed the largest improvements, while American and New American restaurants, which share overlapping vocabulary, showed minimal or negative improvement. This goes to show that food related reviews capture highly informative, cuisine specific signals, making them particularly powerful features for cuisine classification.

6.3 Feature Importance Analysis

Examination of the top predictive features indicates that the food only model learned cuisine related terminology more effectively, which is consistent with our expectations. For instance, the Indian restaurant classifier assigned high importance to terms such as “*naan*”, “*masala*”, and “*biryani*”, while the Mexican classifier emphasized “*tacos*”, “*salsa*”, and “*margaritas*”, as seen in [Figure 14](#). In contrast, the all-reviews model placed greater weight on restaurant names, especially in the fast

food category and more generic terms. Notably, it even identified “*pool*” as the most important feature for the Bar and Grill category, as seen in [Figure 15](#).

6.4 Error Analysis

Analysis of prediction confidence revealed that the model exhibits appropriate calibration: correct predictions had an average confidence of 73.6%, while incorrect predictions averaged 46.2% confidence. This suggests the model “knows when it doesn't know”, when lacking clear cuisine signals, it hedges across multiple categories rather than making confident but wrong predictions.

Inspection of correct and incorrect predictions and their top three candidate classes revealed three main error patterns:

1. **Absence of class content:** Reviews discussing only service, ambiance, or facilities (e.g., bathroom complaints) provided no cuisine / category signal, resulting in near random predictions with low confidence.
2. **Overlapping cuisines:** American, New American, and Steak house categories share significant vocabulary, leading to systematic confusion. The confusion matrix seen in [Figure 16](#) shows American restaurant reviews frequently misclassified as Steak house or New American.
3. **Ambiguous dish names:** Generic terms like “burger” or “pasta” appear across multiple cuisines, reducing discriminative power for categories where these are signature dishes.

The final food only model achieved 66.7% accuracy across 19 classes, demonstrating that review text contains substantial predictive signals for cuisine classification, particularly when filtered to food related content.

We also wanted to see how close our model was when it was getting the classifications wrong, we found that in 19.9% of cases where the model missed the top spot, the correct label was still among its top three choices. This brings the cumulative Top-3 accuracy to 86.6%, confirming that even when the model is ‘wrong,’ it is usually still close to the correct classification, with only 13.4% of reviews being completely missed, as seen in [Figure 17](#).

7. Co-Occurrence Analysis

To further investigate the structure of cuisine-specific food terminology beyond classification performance, we conducted a food co-occurrence analysis using the food items identified by the Named Entity Recognition (NER) model. While the classification models demonstrated that food mentions are highly informative for predicting cuisine type, co-occurrence analysis provides complementary insights into how foods are jointly discussed, revealing shared patterns across cuisines.

7.1 Methodology

The first step was to clean the food column obtained from NER by removing all punctuation as well as case differences. After that we used the inflect library to transform all the plural nouns into their singular form. Food co-occurrence counts were then calculated by identifying all unique food pairs appearing within the same review. Computation at the review level allowed us to avoid inflation from repeated mentions within a single document. To reduce noise only food pairs exceeding a minimum co-occurrence threshold were retained.

7.2 Global Co-Occurrence Trends

Across all cuisines, the strongest food co-occurrences reflect both popularity and cross cuisine generality, as seen on [Figure 18](#). High frequency global pairings include: *burger + fries*, *pizza + salad*, *chicken + rice*, and *chicken + fries*. These combinations align with what we'd expect to see and appear across multiple restaurant categories, explaining their reduced power in classification tasks.

Analysis of the top 30 most mentioned foods further highlights this phenomenon. Items such as *pizza*, *burger*, *chicken*, *fries*, and *salad* dominate the corpus, suggesting that while these foods are central to consumer discourse, they are not necessarily cuisine-specific.

7.3 Cuisine-Specific Co-Occurrence Patterns

The resulting co-occurrence profiles reveal highly interpretable and cuisine-consistent food pairings across all 19 cuisine categories.

Mexican restaurants are characterized by strong pairings such as *chips + salsa*, *beans + rice*, and *tacos + salsa*, reflecting canonical meal structures and shared side dishes. Fast Food and Hamburger restaurants display the strongest overall co-occurrence signals in the dataset, dominated by *burger + fries*, which is also the single strongest food pair globally with 376 total co-occurrences.

More specialized cuisines demonstrate lower-frequency but distinct structures. Indian restaurants are characterized by pairings such as *naan + rice*, *butter chicken + naan*, and *chicken tikka masala + garlic naan*, highlighting the role of bread combinations. Japanese restaurants show strong associations between *sushi*, *rolls*, *hibachi*, and *rice* reinforcing culturally specific shared dishes.

7.4 Network Structure and Cuisine Distinctiveness

The food network visualizations ([Figure 19](#)) reveal meaningful differences in structural cohesion across cuisines. Hamburger and Pizza cuisines show the densest connectivity, with "fry" and "pizza" serving as central hub nodes respectively,

suggesting these items frequently appear together with numerous other menu items in reviews. In contrast, Mediterranean, Indian, and Thai cuisines exhibit sparser, more specialized networks with fewer interconnections between dishes.

Notably, the restaurants that are dominated by cuisine-specific food combinations (such as Indian, Japanese and Thai) are the same ones that achieved the largest performance in the food-only classification model. This alignment suggests that distinctive co-occurrence structure, not just individual food frequency, plays a key role in enabling robust cuisine classification.

7.5 Co-Occurrence Summary

Overall, the co-occurrence analysis reinforces and extends the findings from the classification task. While individual food mentions provide strong predictive signals, the structure of food co-occurrence reveals deeper culinary organization that explains both model successes and failures. The alignment between cuisine-specific networks, classification performance, and confusion patterns suggests that food co-occurrence networks capture meaningful representations of cuisine identity, offering a valuable tool for both exploratory analysis and feature engineering in the future.

8. Conclusion

This project demonstrates that restaurant reviews contain substantially more valuable information than just star ratings alone. While numerical scores provide a convenient summary of customer satisfaction, our analysis shows that the textual content of reviews captures deeper, more nuanced information about both customer satisfaction and restaurant identity. Through sentiment analysis, we find a strong relationship between review polarity and star ratings, but also reveal systematic differences where the text explains dissatisfaction in ways a single score cannot. In parallel, cuisine classification results show that review text can also be used to identify specific dishes and even the cuisine types of a restaurant. Overall, these findings confirm that user-generated text is not just a justification for a score, but a richer and more expressive source of information that allows text mining models to uncover insights about customer experience and restaurant characteristics.

References

- [1] O. Bulkrock, A. Qusef and A. BaniMustafa, "Sentiment Analysis of Customer Feedback and Reviews in E-Commerce Systems," 2025 International Conference on New Trends in Computing Sciences (ICTCS), Amman, Jordan, 2025, pp. 379-385, doi: 10.1109/ICTCS65341.2025.10989450.
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- [4] J. C. Sugitomo, N. Kevin, N. Jannatri and D. Suhartono, "Sentiment Analysis using SVM and Naïve Bayes Classifiers on Restaurant Review Dataset," 2021 1st International Conference on Computer Science and Artificial Intelligence (ICCSAI), Jakarta, Indonesia, 2021, pp. 100-108, doi: 10.1109/ICCSAI53272.2021.9609776.

Figures

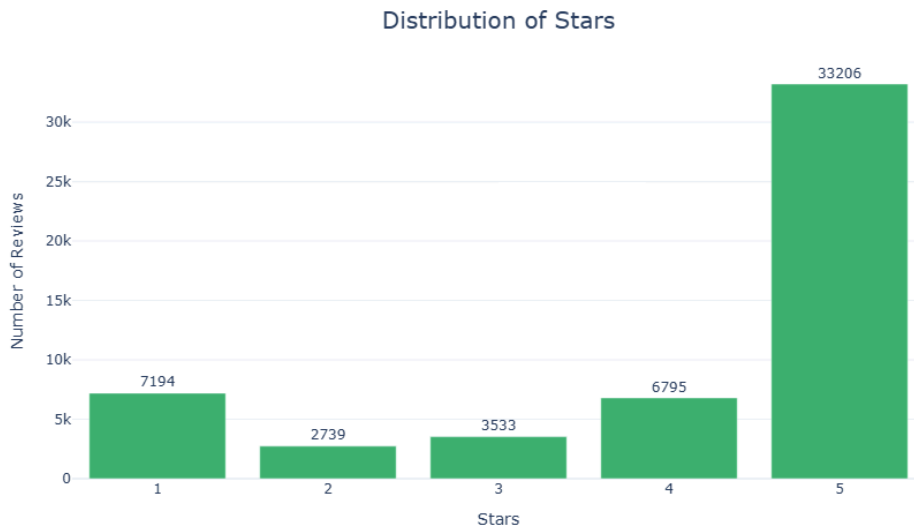


Figure 1: Distribution of 'Stars' rating

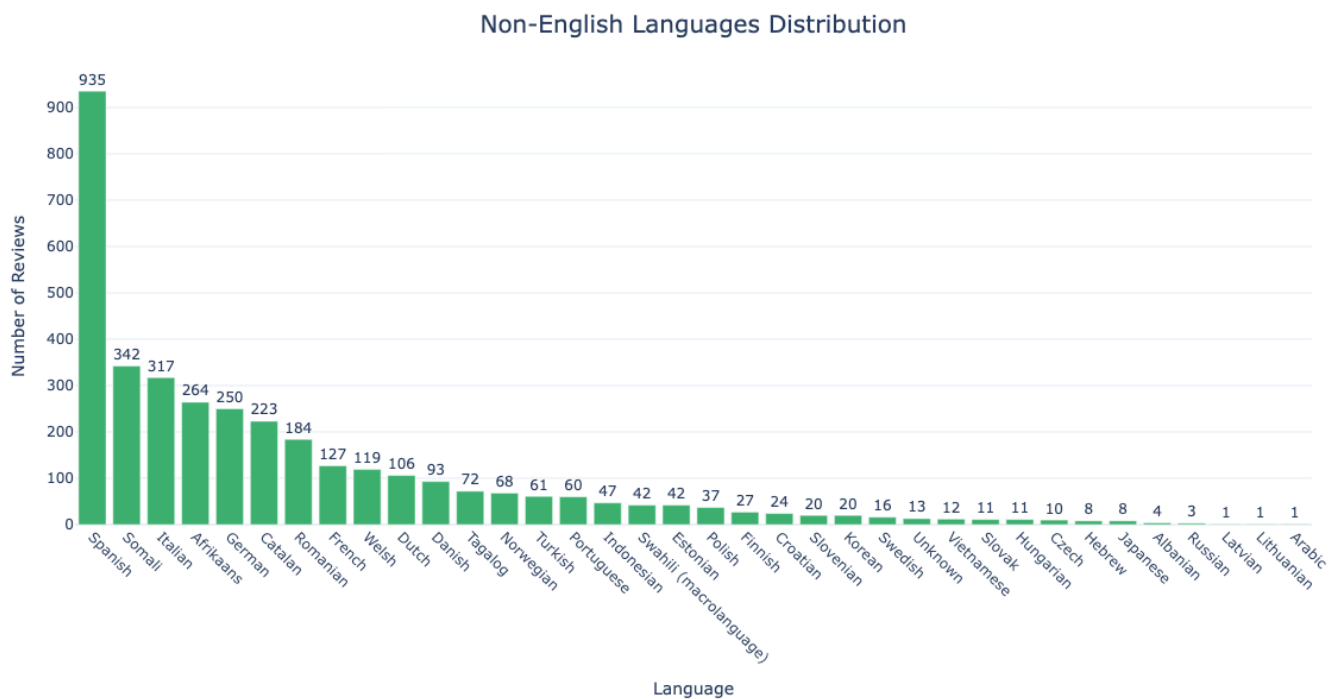


Figure 2: Distribution of non english review languages

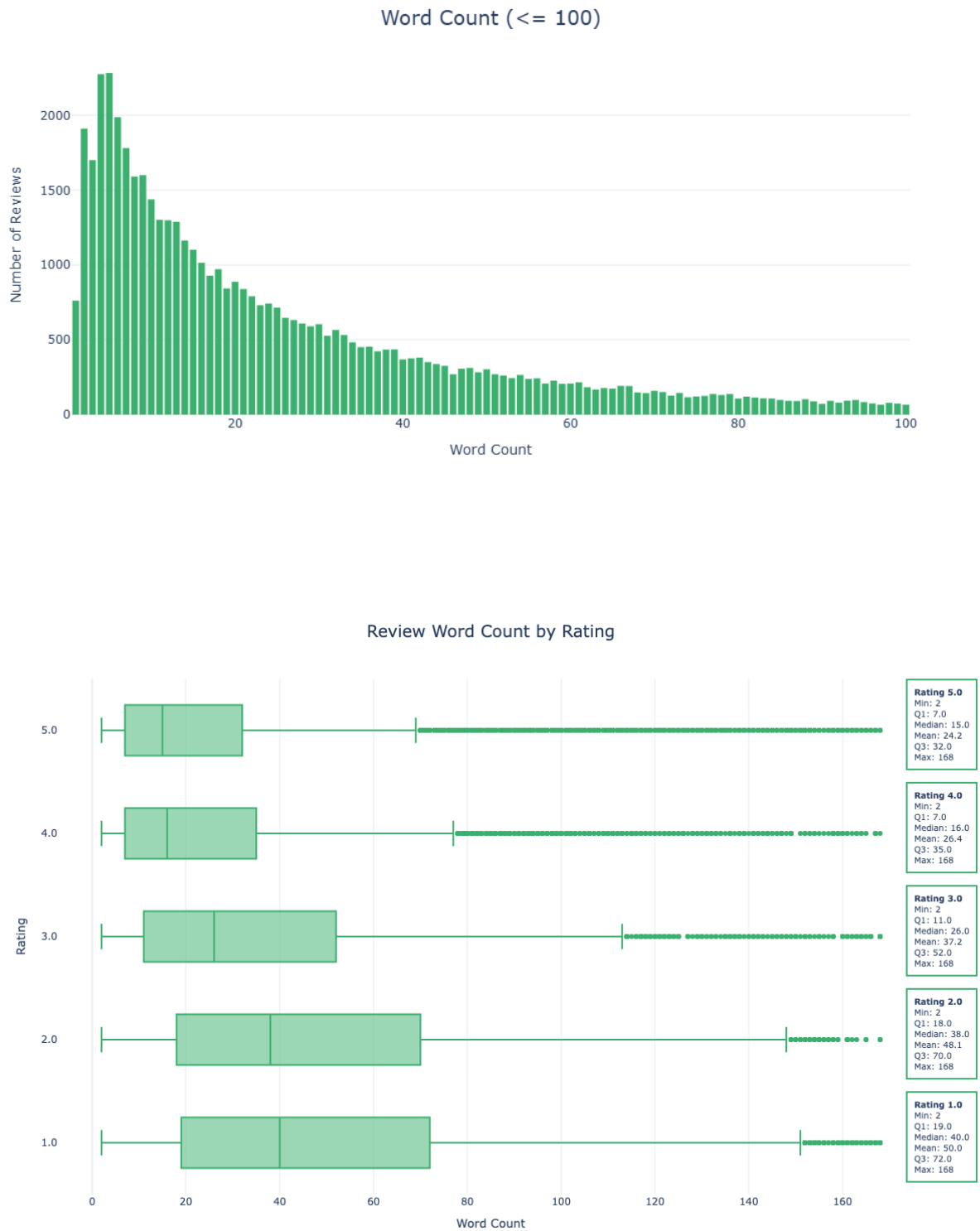


Figure 3: Word count distribution and rating distribution. (a) Word count distribution
(b) Rating word count distribution

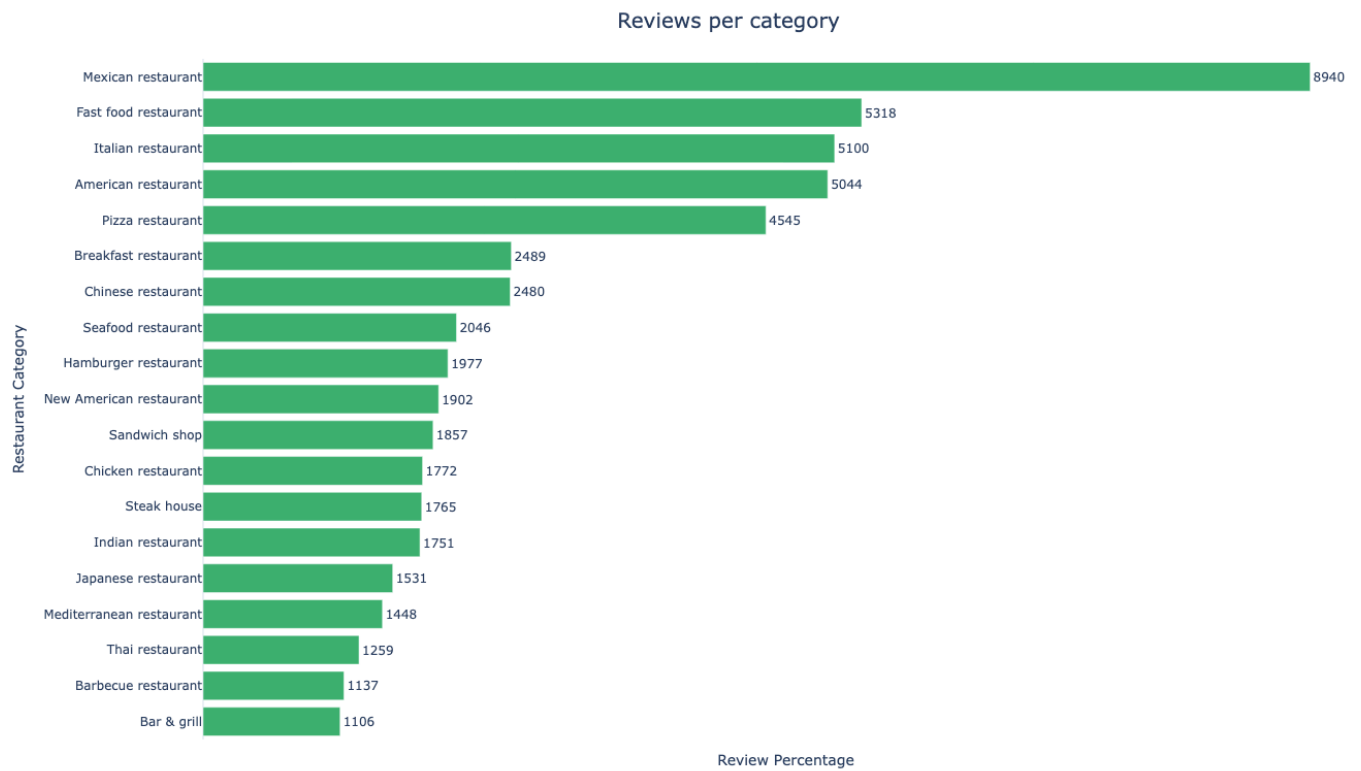


Figure 5: Cuisine category distribution

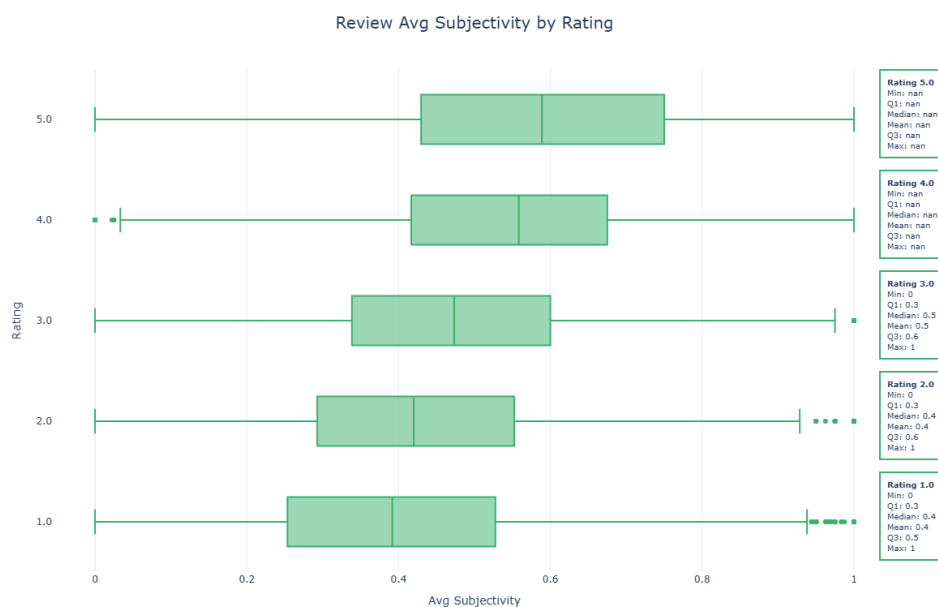


Figure 6: TextBlob Subjectivity boxplots across true stars

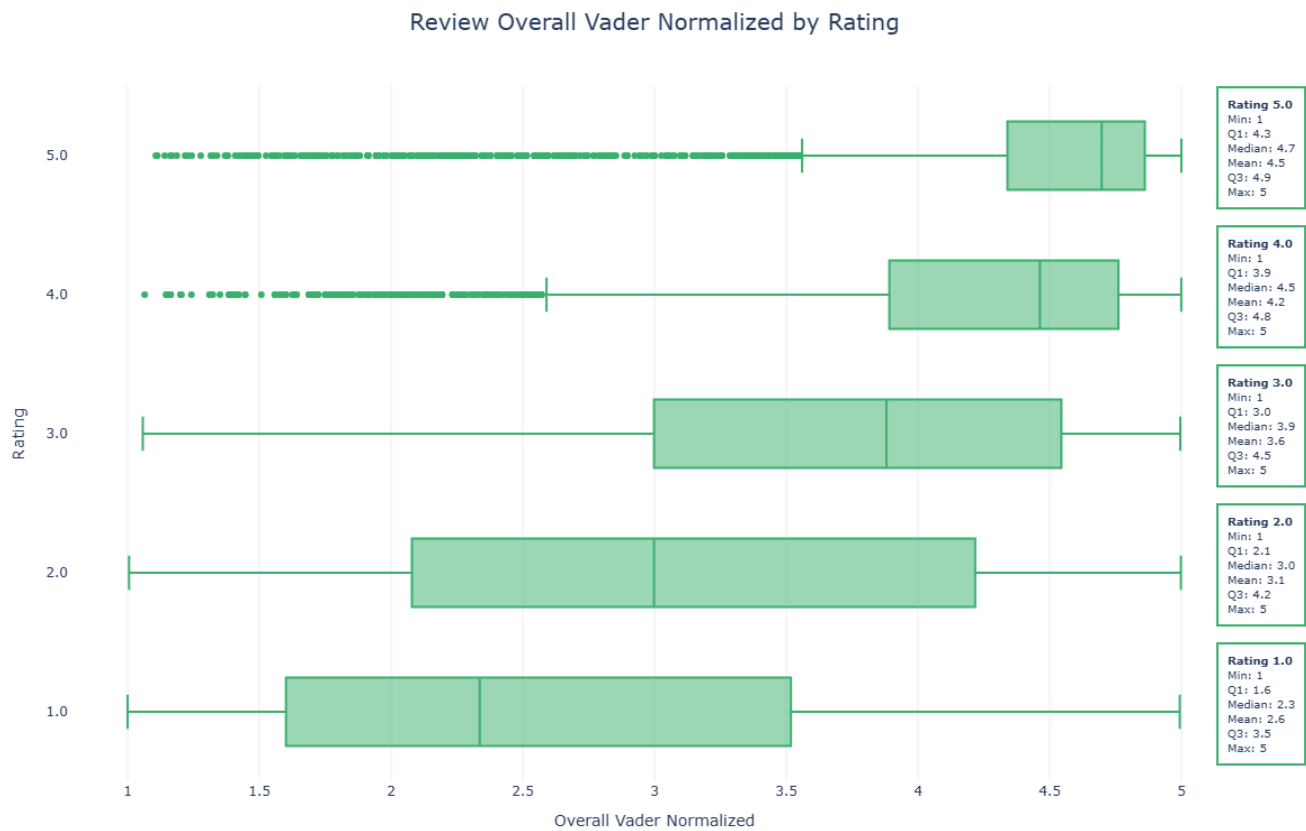


Figure 7: Vader overall scores across true stars

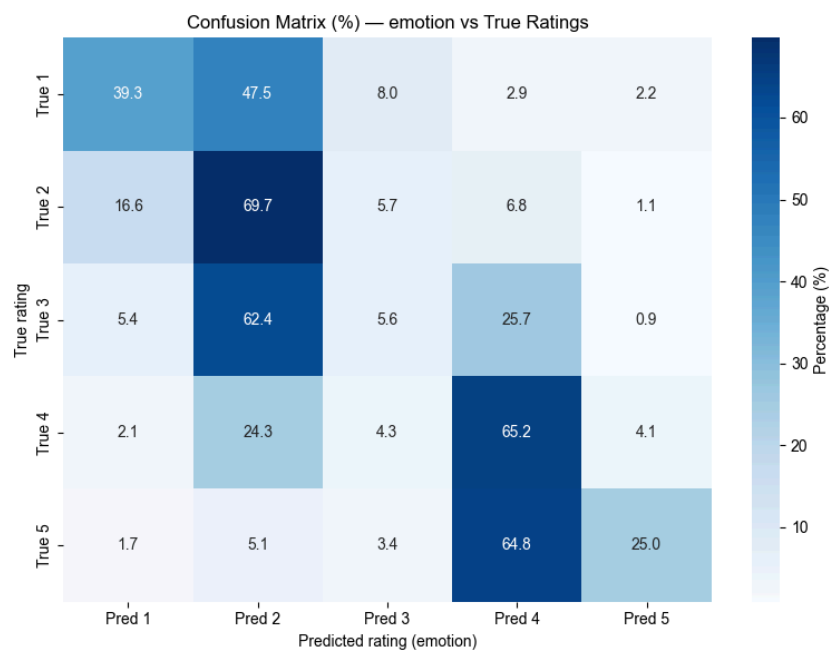


Figure 8: Confusion Matrix of pre-trained model emotion with true stars



Figure 9: Positive / Negative wordclouds by VADER compound scores

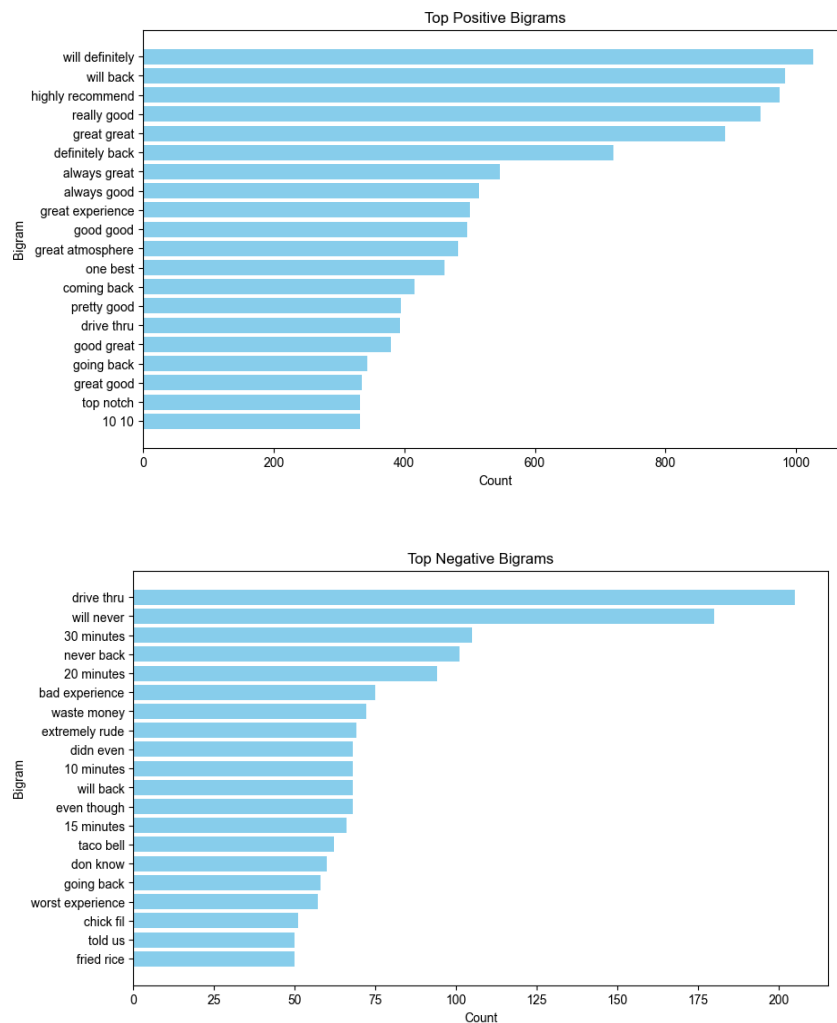


Figure 10 (A) (B): Most frequent bigrams in positive / negative reviews

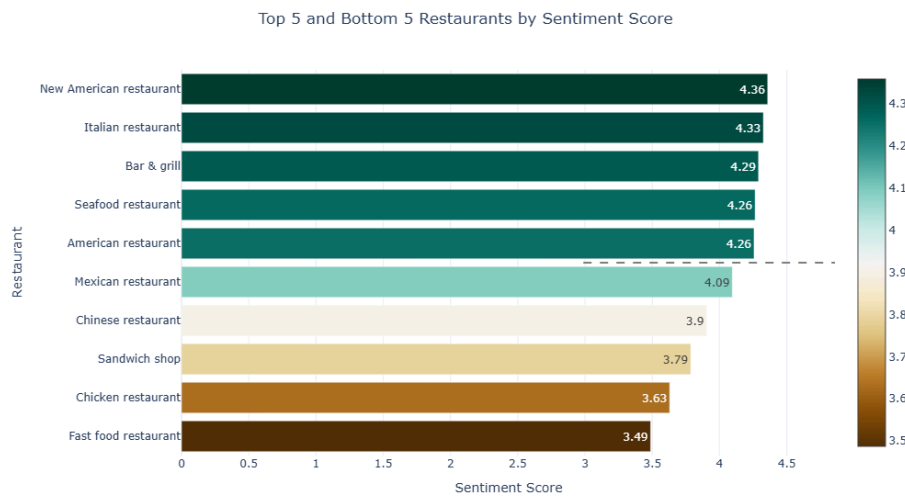


Figure 11: Better / Worst reviewed restaurants according to VADER

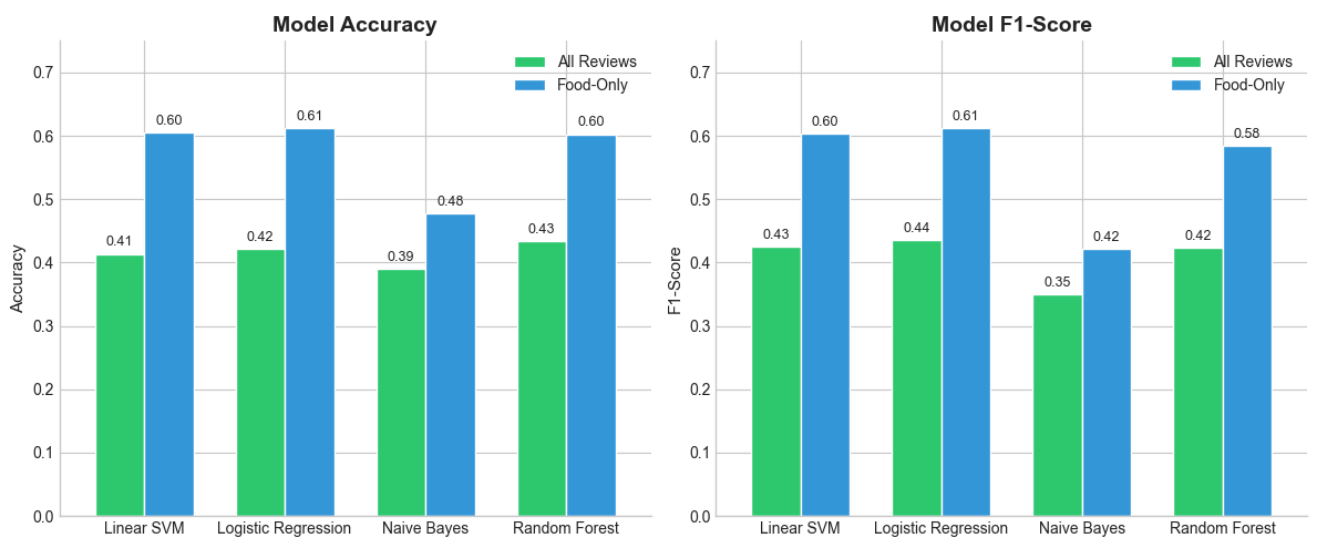


Figure 12: Cuisine classification model performance comparison

	Cuisine	F1 (All)	F1 (Food)	F1 Δ	Support (All)	Support (Food)
0	Indian restaurant	0.582	0.882	+0.300	339	152
1	Mediterranean restaurant	0.413	0.695	+0.282	275	123
2	Thai restaurant	0.421	0.683	+0.262	244	105
3	Japanese restaurant	0.444	0.695	+0.251	293	143
4	Barbecue restaurant	0.483	0.720	+0.237	218	127
5	Sandwich shop	0.504	0.739	+0.234	343	158
6	Breakfast restaurant	0.396	0.621	+0.225	467	172
7	Seafood restaurant	0.400	0.622	+0.222	390	199
8	Chinese restaurant	0.436	0.639	+0.204	469	190
9	Mexican restaurant	0.635	0.826	+0.191	1696	706
10	Hamburger restaurant	0.530	0.692	+0.161	380	260
11	Pizza restaurant	0.665	0.817	+0.151	860	563
12	Italian restaurant	0.516	0.667	+0.151	982	470
13	Bar & grill	0.247	0.378	+0.131	210	92
14	Chicken restaurant	0.410	0.527	+0.116	330	172
15	American restaurant	0.348	0.457	+0.109	972	409
16	Steak house	0.389	0.477	+0.088	345	132
17	Fast food restaurant	0.532	0.615	+0.083	986	368
18	New American restaurant	0.325	0.318	-0.007	368	153

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SUMMARY

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Average F1 improvement: +0.179

Best improvement: +0.300 (Indian restaurant)

Worst improvement: +-0.007 (New American restaurant)

Cuisines improved: 18 / 19

Figure 13: Final model scores F1 score comparison across both datasets

Top Predictive Features by Cuisine (Food-Only Model)

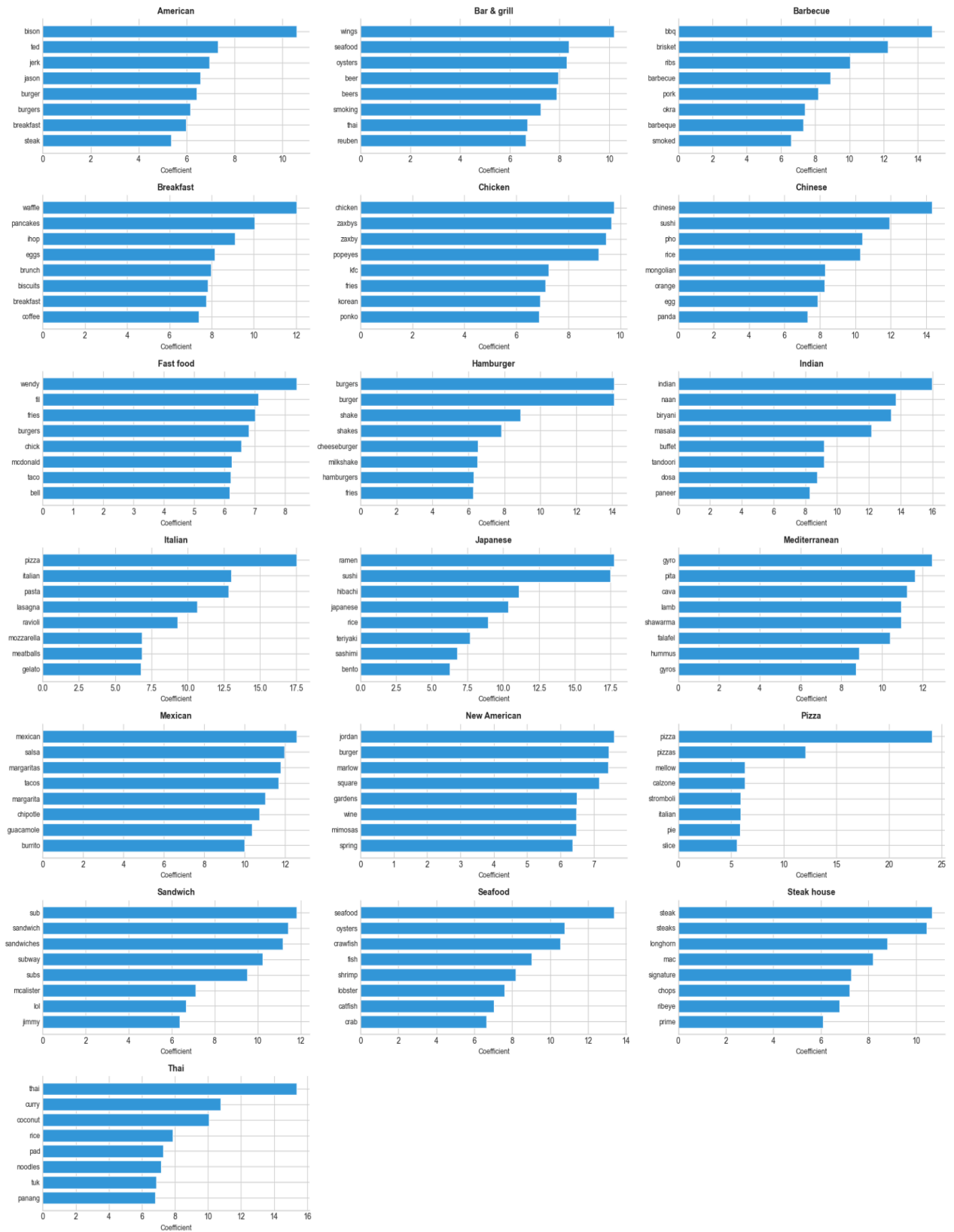


Figure 14: Feature importances of food only model

Top Predictive Features by Cuisine (All Reviews Model)

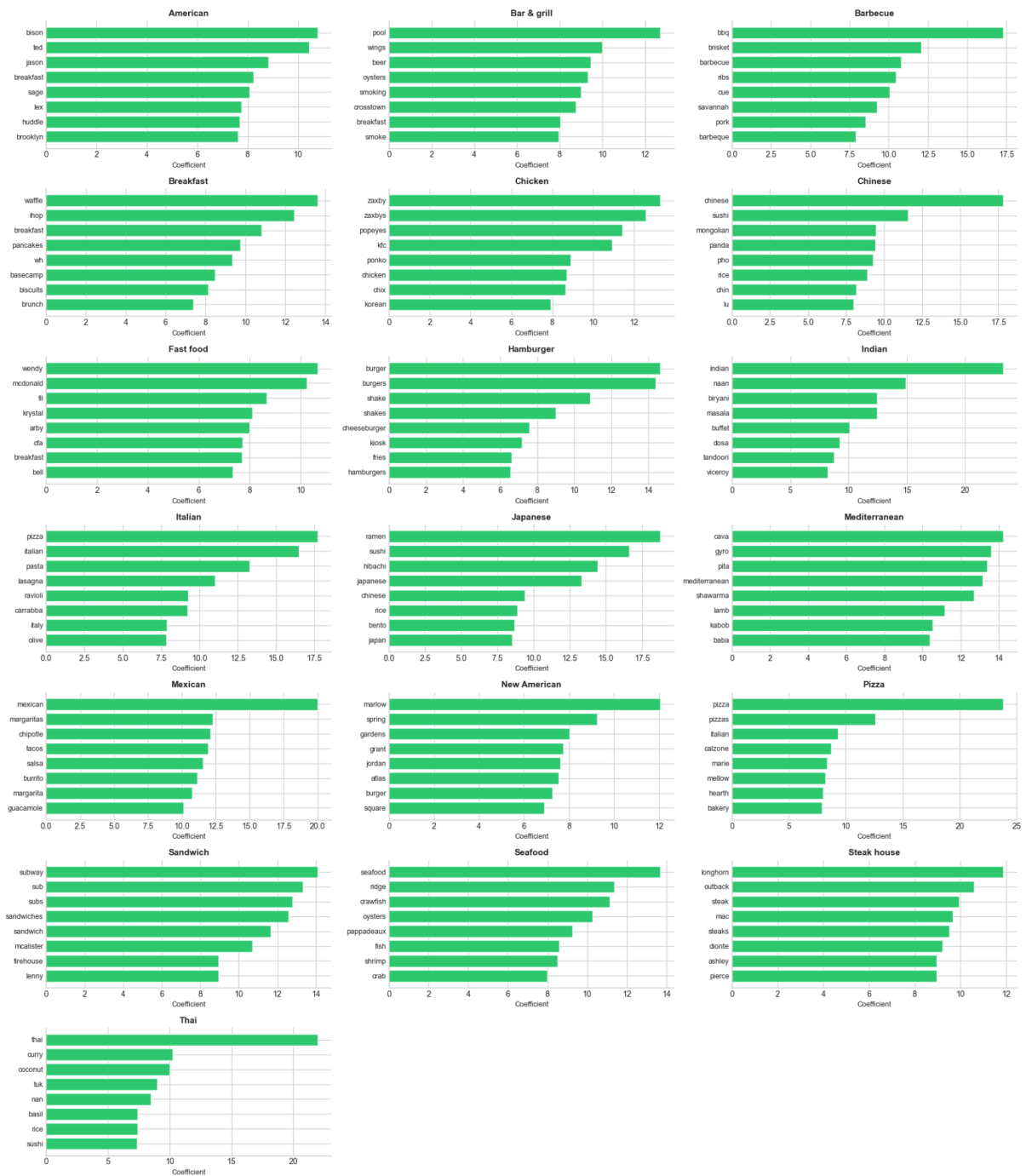


Figure 15: Feature importances of all reviews model

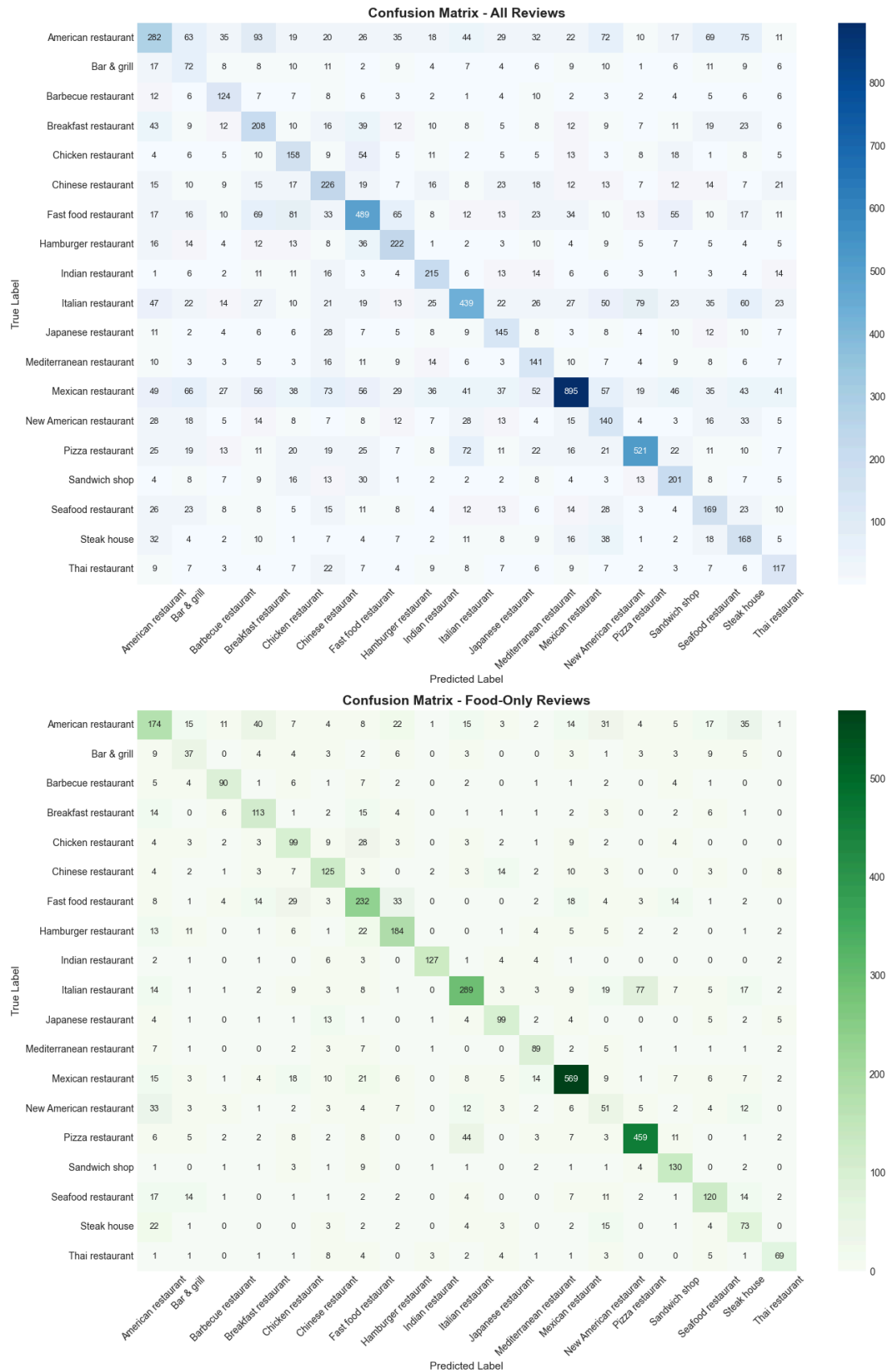


Figure 16: Confusion matrices for both models

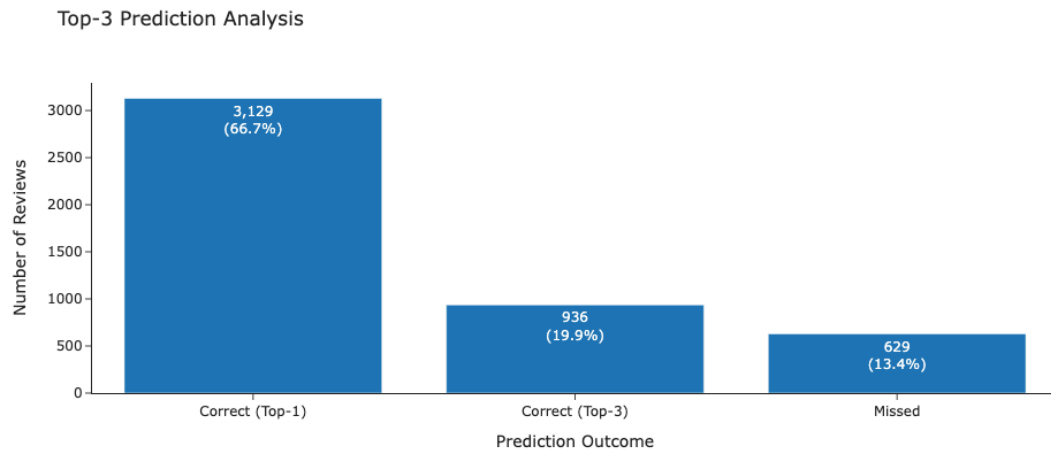


Figure 17: comparing top-1 vs. top-3 model success rate

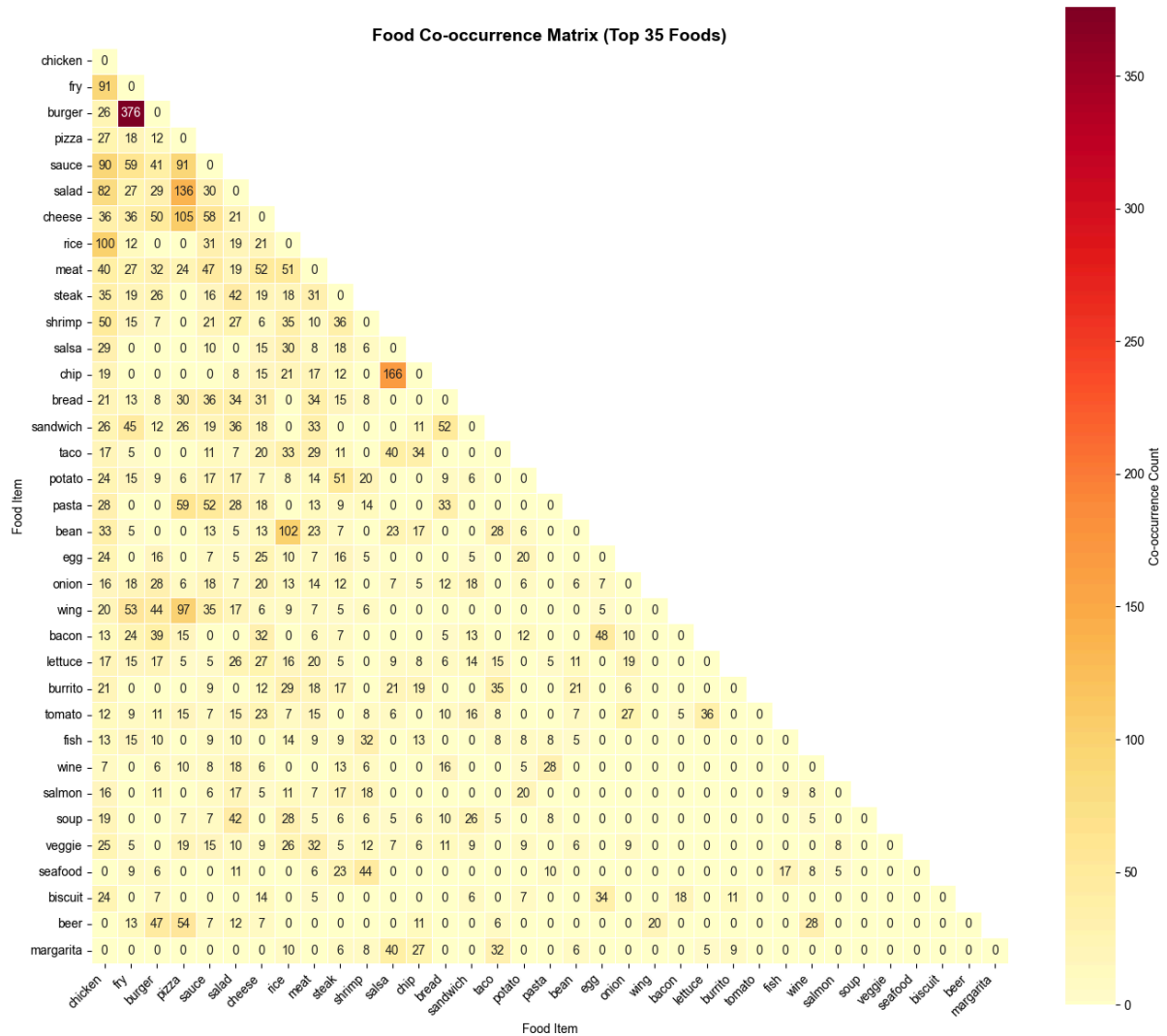


Figure 18: Food Co-occurrence Matrix

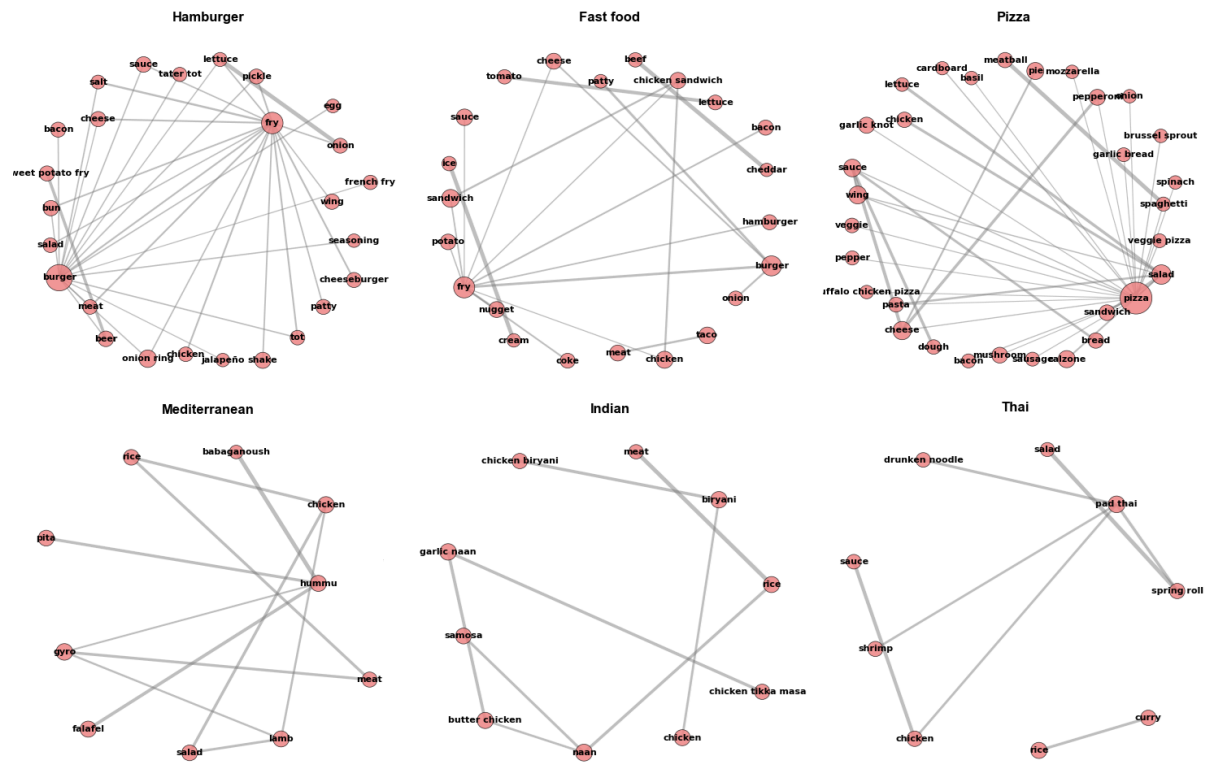


Figure 19: Food Co-occurrence Networks by Cuisine Type